

Table 8: Performance comparison of StreamingLLM against the default truncation baseline in LongBench (Bai et al., 2023). The baseline truncates inputs to 1750 initial and 1750 final tokens. StreamingLLM 4+3496 uses 4 attention sink tokens and 3496 recent tokens, while StreamingLLM 1750+1750 uses 1750 tokens for both initial and recent segments.

Llama2-7B-chat	Single-Document QA		Multi-Document QA		Summarization	
	NarrativeQA	Qasper	HotpotQA	2WikiMQA	GovReport	MultiNews
Truncation 1750+1750	18.7	19.2	25.4	32.8	27.3	25.8
StreamingLLM 4+3496	11.6	16.9	21.6	28.2	23.9	25.5
StreamingLLM 1750+1750	18.2	19.7	24.9	32.0	26.3	25.9

D LONG-RANGE BENCHMARK EVALUATION

We evaluated StreamingLLM using the Llama-2-7B-chat model (max context length 4k) on LongBench (Bai et al., 2023), which encompasses three key NLP tasks: single-document QA (NarrativeQA (Kočiský et al., 2017) and Qasper (Dasigi et al., 2021)), multi-document QA (HotpotQA (Yang et al., 2018) and 2WikiMQA Ho et al. (2020)), and summarization (GovReport (Huang et al., 2021), MultiNews (Fabbri et al., 2019)). LongBench sets a default max sequence length of 3,500 tokens for the Llama-2-7B-chat model, truncating from the middle to preserve beginning and end information (1,750 tokens each). Table 8 shows that StreamingLLM with a 4+3496 cache configuration underperforms compared to the truncation baseline, likely due to the loss of crucial initial input prompt information. However, aligning the attention sink number to 1750 restores performance to the level of the text truncation baseline. These results corroborate the findings in Section C, demonstrating that StreamingLLM’s effectiveness is contingent on the information within its cache, with in-cache performance comparable to the text truncation baseline.

E LLAMA-2-7B ATTENTION VISUALIZATION ON LONGER SEQUENCES

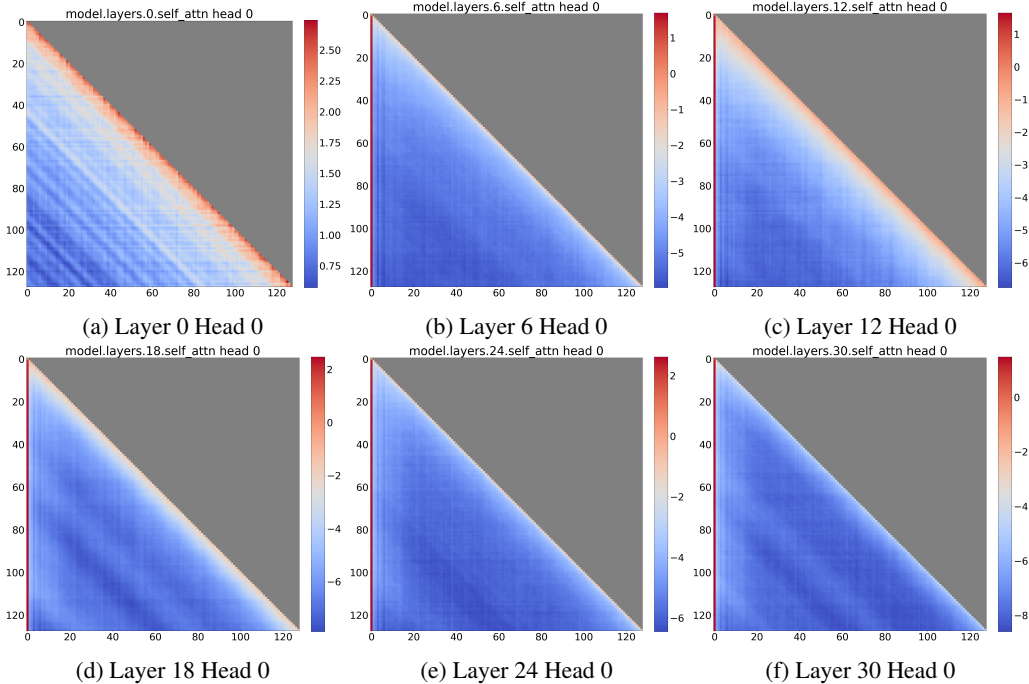


Figure 11: Visualization of the *average* attention logits in Llama-2-7B over 256 sentences, each with a length of 128.

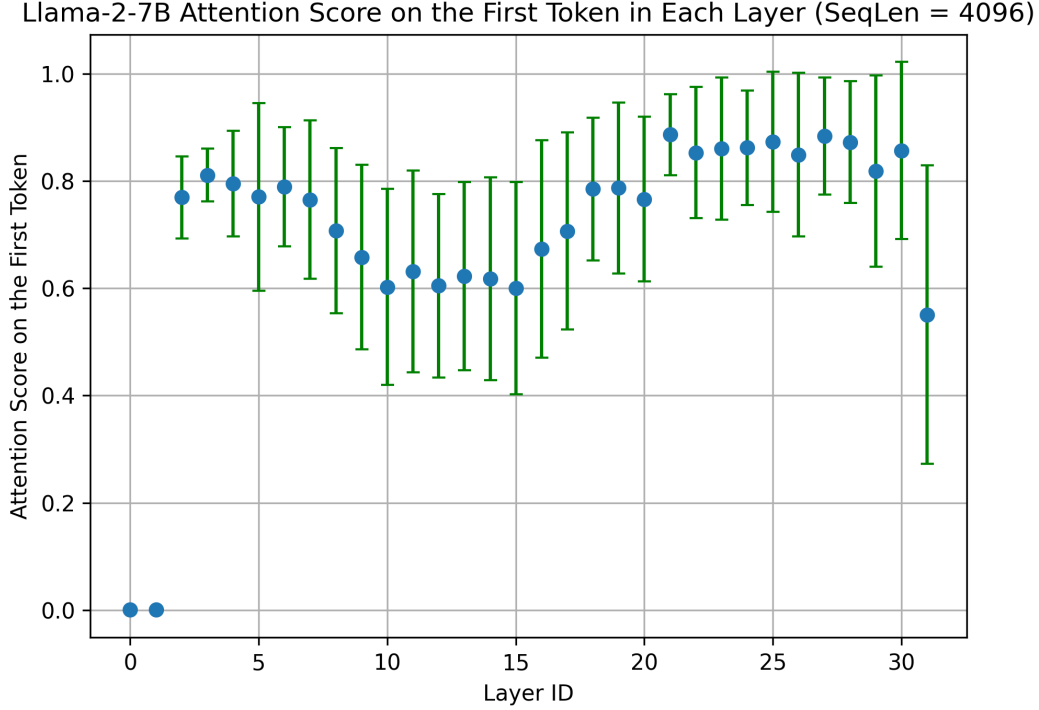


Figure 12: Visualization of attention scores (after SoftMax) on the first token across layers in Llama-2-7B. Attention Scores are the 4096th token’s attention towards the first token in each layer. The error bars are the standard deviation of the first token’s attention scores across different heads in one layer. Results are averaged over 256 sentences, each having a length of 4096 tokens.

Figure 2 visualizes the attention map of Llama-2-7B using short sequences (length of 16) for clarity. We further visualize the attention of Llama-2-7B on longer sequences (length of 128) in Figure 11. We find the observations on short sequences also hold on longer sequences, where the attention scores of the initial tokens are much higher than the rest of the tokens in most layers, regardless of the distance between the initial tokens and the tokens in the rest of the sequence. Because the longer the sequence, the thinner the attention sinks’ scores are visualized on the heatmap. We further analyze the attention distribution on longer sequences (length of 4096) using a different method in Section F.

F QUATITATIVE ANALYSIS OF ATTENTION SINKS IN LONG INPUTS

Figures 2 and 13 illustrate the attention sink phenomenon using short sequences for clarity. Extending this analysis, Figure 12 demonstrates the distribution of attention scores (after SoftMax) towards the first token in lengthy inputs (sequence length of 4096). We average attention scores across 256 sequences, with each sequence comprising 4096 tokens. The plotted data represent the attention allocated by the 4096th token to the initial token in every layer. Notably, the attention scores for the first token are significantly high, often exceeding half of the total attention, except for the two bottom layers. This observation empirically substantiates the preferential focus on the first token by the majority of layers and heads, irrespective of other tokens’ distances within the sequence. Such a trend underscores the critical role of the initial tokens in a sequence, as their removal has a huge impact on language model performance due to a large portion of the denominator in the SoftMax function being removed.

G LLAMA-2-70B ATTENTION VISUALIZATION

Figure 2 shows the attention visualization of Llama-2-7B, we further visualize the attention of Llama-2-70B in Figure 13. We find the observation on Llama-2-7B also holds on Llama-2-70B,

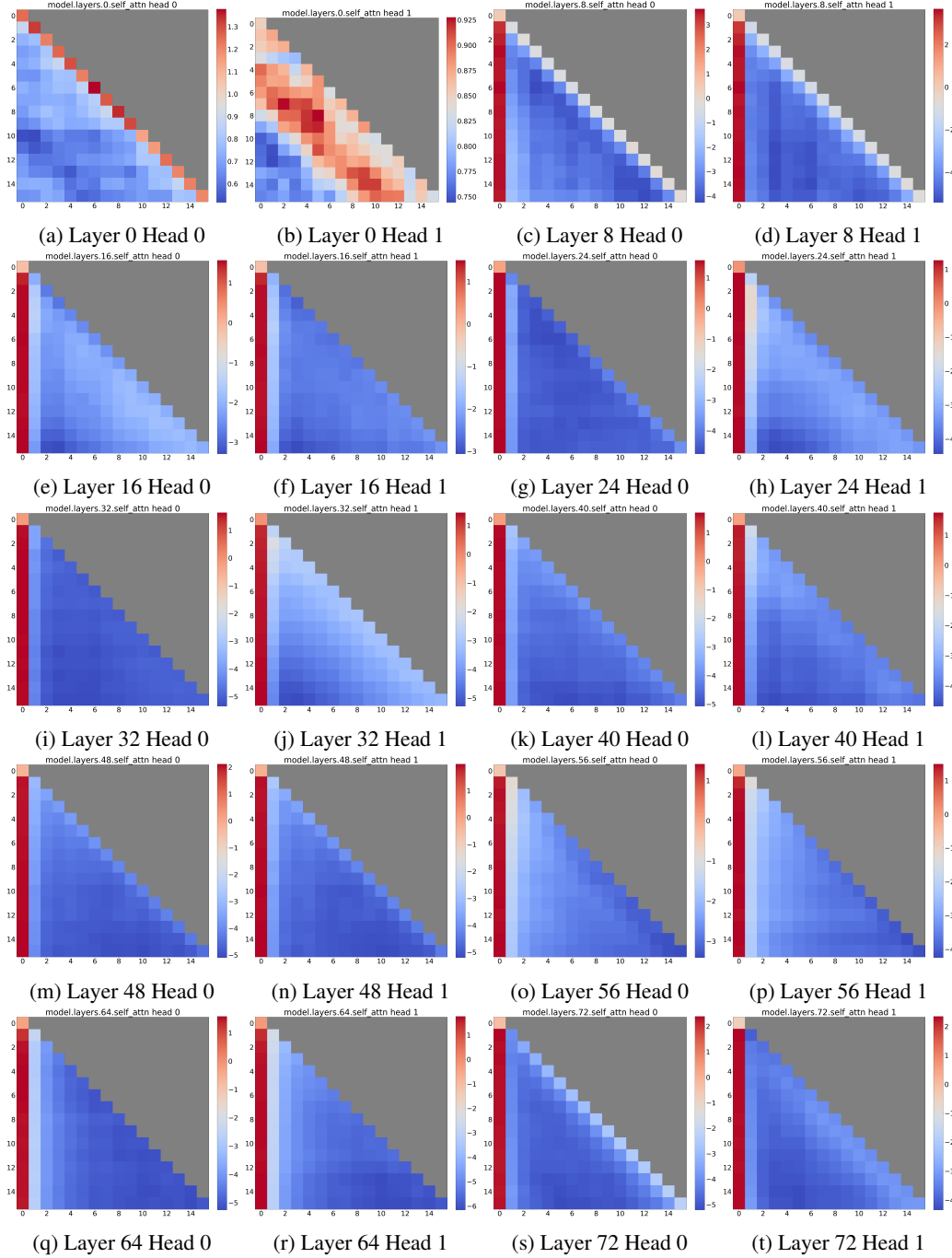


Figure 13: Visualization of the *average* attention logits in Llama-2-70B over 256 sentences, each with a length of 16.

where the attention scores of the initial tokens are much higher than the rest of the tokens in most layers.

H ATTENTION SINKS IN ENCODER TRANSFORMERS

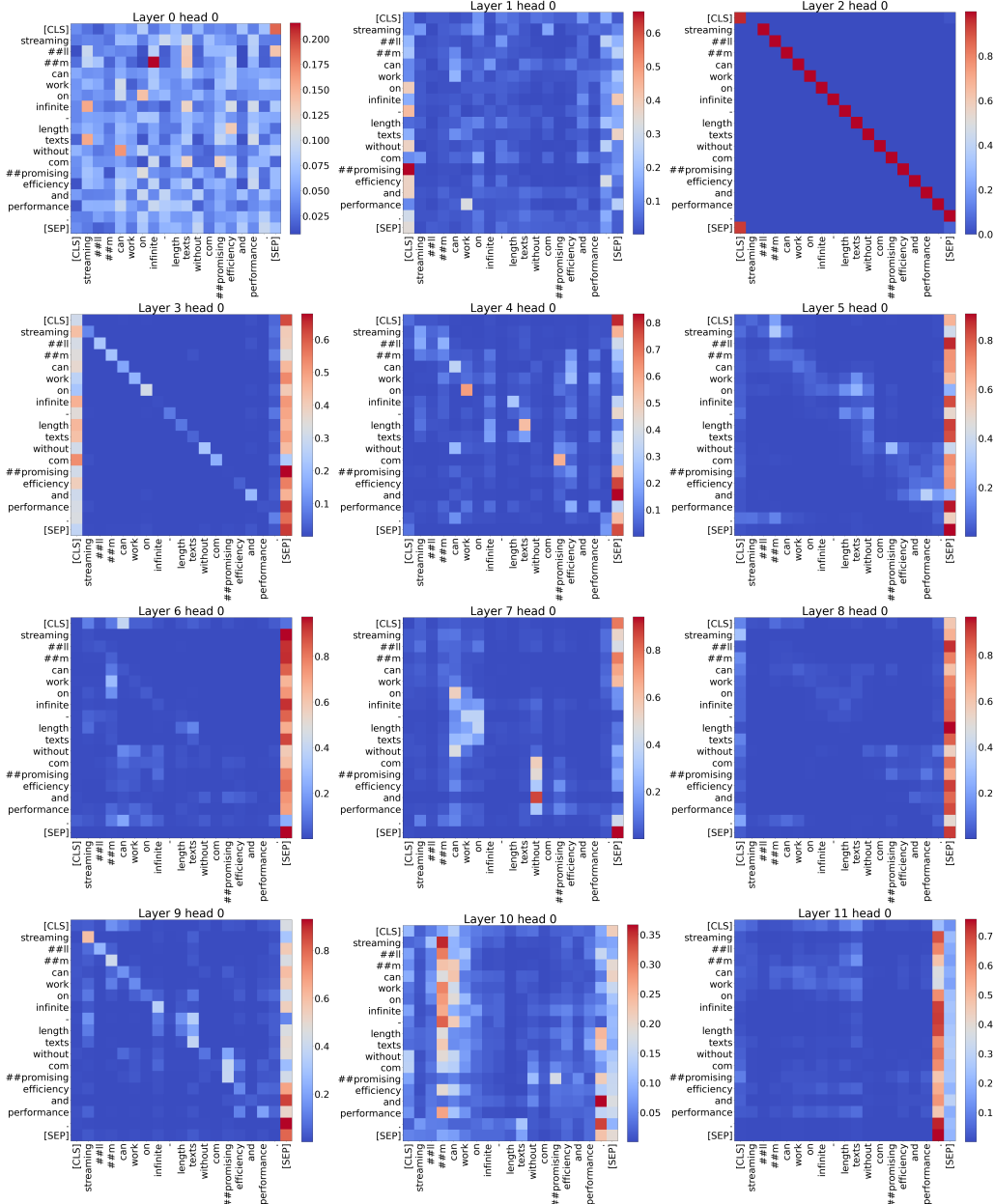


Figure 14: Visualization of attention maps for sentence "StreamingLLM can work on infinite-length texts without compromising efficiency and performance." in BERT-base-uncased.

In this paper, we mainly explore the attention sink phenomenon observed in autoregressive, decoder-only language models like GPT and Llama. Building upon the insights from Section 3.1, we propose that this phenomenon likely extends to other Transformer architectures, including encoder models such as BERT (Devlin et al., 2019) and ViT (Dosovitskiy et al., 2021). This assumption stems from the fact that these models share a similar Transformer structure and utilize SoftMax attention mechanisms. To substantiate our hypothesis, we analyze the attention patterns of BERT-base-uncased,